Jay Patrikar

Pittsburgh, PA 15217

✉ jaypat@cmu.edu +1 (412) 692-1516

EDUCATION

Carnegie Mellon University | School Of Computer Science (Robotics Institute), *Pittsburgh, PA*
Master of Science in Robotics — *Fall 2018 – Summer 2020*[1]
GPA: 4.08 / 4.00

Indian Institute of Technology Kanpur, *Kanpur, India*
Masters and Bachelors of Technology (Intergrated) in Aerospace Engineering — *Fall 2013 – Summer 2018*
CPI (PG) : 10.0/10.0, CPI (UG): 8.6/10.0
Minors : Controls Systems (EE) and English Literature (Humanities)

Selected Courses: Planning and Decision-making in Robotics, Deep Reinforcement Learning, Computer Vision, Introduction to Machine Learning (PhD), Flight Mechanics, Robot Localization and Mapping[1]

RESEARCH EXPERIENCE

Airlab, *Carnegie Mellon University*
Guide: Dr. Sebastian Scherer — *2018 – Now*
Thesis Title: Kinodynamic Path Planning for Improving Autonomous Aerial Mobility in Urban Areas[1]

- Prototyping a real-time kinodynamic learning-informed path planning algorithm with curvature and curvature-rate constraints using a two-point boundary value problem formulation.
- Carried out a flight data collection campaign to collect energy consumption data for a done-based system to train a first-principle physics-based energy model.

Intelligent Guidance & Control Laboratory, *Indian Institute of Technology Kanpur*
Guide: Dr. Mangal Kothari — *2016 – 2018*
Thesis Title: A 3D Guidance Law for Path Following of Unmanned Aerial Vehicles

- Developed and flight tested a novel guidance law for 3D path following of fixed-wing Unmanned Aerial Vehicles using input-based nested saturation techniques for straight line and circular geometries.
- The proposed law was extended to spline-based trajectories for sequential landing of multiple fixed wing UAVs.

Unmanned Aerial Vehicles Laboratory, *Indian Institute of Technology Kanpur*
Guide: Dr. A.K.Ghosh — *2015 – 2018*

- Designed and tested autopilot systems for fixed wing and rotary UAVs with applications in surveillance and target-tracking.
- Research areas: Flight Mechanics, RTOS, Systems Integration, System Identification, Flight Parameter Estimation, Vision-based tracking

National Aerothermochemistry and Hypersonics Laboratory, *Texas A&M University*
Guide: Dr. R Bowersox — *Summer 2016*

- Studied Boundary Layer Transition on Elliptic cone geometries in Hypersonic flow using thermal flow visualization.
- Also worked on the construction of the now operational hypersonic expansion tunnel (HXT) for high enthalpy research.
- Research areas: Hypersonics, Boundary layer transitions, Wind tunnel modelling.

PUBLICATIONS

Kinodynamic Wind-Aware Path Planning for Drone Operations in Urban Areas
***J Patrikar**, V Dugar, V Arcot, S Scherer* — *2020*
International Conference on Automated Planning and Scheduling (ICAPS) (Submitted)

Development of an Energy Model for Quadcopter Package Delivery Drones[1]
*T Rodrigues, **J Patrikar**, B Wagner, S Scherer, C Samaras* — *2020*
International Journal of Transportation Research Part C: Emerging Technologies

Nested Saturation Based Guidance Law for Unmanned Aerial Vehicles
***J Patrikar**, VR Makkapati, A Pattanaik, H Parwana, M Kothari* — *2019*
Journal of Dynamic Systems, Measurement, and Control, ASME

Sequential Auto-Landing of Multiple UAVs using Control Constrained Path Following
***J Patrikar**, VR Makkapati, M Kothari* — *2019*
AIAA Guidance, Navigation, and Control Conference

[1] Planned

Convolutional Neural Network Based Sensors for Mobile Robot Relocalization
J Patrikar*, *H Sinha**, *EG Dhekane**, *G Pandey, M Kothari* *2018*
23rd International Conference on Methods & Models in Automation & Robotics

A novel fully quaternion based nonlinear attitude and position controller
*H Parwana, **J Patrikar**, M Kothari* *2018*
AIAA Guidance, Navigation, and Control Conference

A Low-Cost Tilt-Augmented Quadrotor Helicopter : Modeling and Control
*M Bhargavapuri,**J Patrikar**, SR Sahoo, M Kothari* *2018*
International Conference on Unmanned Aircraft Systems (ICUAS)

MARAAL: A Low Altitude Long Endurance Solar Powered UAV For Surveillance and Mapping Applications
*VS Dwivedi, **J Patrikar**, A Addamane, AK Ghosh* *2018*
23rd International Conference on Methods & Models in Automation & Robotics

INDUSTRY COLLABORATION PROJECTS

Development of a Preflight Path Planning Tool for General Aviation Pilots
Near Earth Autonomy Inc., Pittsburgh, PA *2019*
- Developing a multi-objective risk-aware graph-based pre-flight planner for VFR (Visual-Flight Rules) cross-country flights
- Areas: GIS, Database Management, Graph-based planning, risk modelling

Development of UAV Dynamics Model and Study of Control Effectiveness
Centre for High Energy Systems and Sciences (CHESS), Hyderabad, India *2018*
- Designed a full system dynamics simulator for fixed-wing UAVs and performed ablation studies to characterise control surface effectiveness.
- Areas: 6-DOF Simulations, Control Ablation Studies

Development of a Autonomous Payload Delivery System
Delhivery Couriers Pvt Ltd, Delhi, India *2017*
- Prototyping and flight testing of a multi-rotor platform to deliver a 1kg package over a distance of 3km.
- Areas: Systems integration, Flight Testing, Product design

Development of a High Altitude Surveillance Platform
TATA Advanced Systems Pvt Ltd, Banglore, India *2016*
- Prototyping and flight testing of a high-altitude fixed wing platform for operations at $> 4500m$ MSL.
- Areas: High Altitude Flight Mechanics, Design Optimization, Parameter Estimation, UAV Operations Management

AWARDS & RECOGNITION

- **2 A+ grades** for exceptional performance in courses at Carnegie Mellon University
- **2 Academic Excellance Awards** (for distinctive academic performance in years 2016-17 and 2017-18) at Indian Institute of Technology Kanpur
- Awarded the **Summer Undergraduate Research Grant for Excellence (SURGE)** for Summer 2015
- **4 A+ grades** for exceptional performance in courses at Indian Institute of Technology Kanpur
- **All India Rank 2219** (amongst 13,00,000 students) in IIT-Joint Entrance Examination 2013

LEADERSHIP AND MENTORING

Society of Aerospace Engineers, IIT Kanpur
President *2016 – 2017*
- Society regularly organises academic activities like organising industrial trips, workshops, seminars and guest lectures for fostering enthusiasm in Aerospace Engineering.

Department Undergraduate Committee (DUGC), Department of Aerospace Engineering , IIT Kanpur
Elected Student Representative *2015 – 2016*
- Elected as a student nominees from the UG department (approx. 450 students) for voicing the opinions of the students in critical matters like course restructuring, program termination appeals etc. and ensuring grievance redressal.

Hostel Executive Committee
Elected Secretary Of Science and Technology *2014 – 2015*
- Led 204 students to victory in Takneek 2014 (series of inter-hostel technical competitions).
- Part of the committee that manages the day to day running of hostel with over 623 residents.

SKILLS & OTHERS

Languages: English, Hindi, Marathi (Native), French (A1 Level Cerification)
Computer Languages: C++, Python, SQL
Tools and Libs: OMPL, ROS, PX4, PostGres, PostGIS, Redis, TensorFlow
Certifications: Remote Pilot Certificate FAA Part 107